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1-B:

The equimarginal principle states that consumers will choose a combination of goods to maximise their total utility. This will occur where the consumer will consider both the marginal utility MU of goods and the price. This equilibrium is expressed as:
$$\text{Marginal Utility of Good X} / \text{Price of X} = \text{Marginal Utility of Good Y} / \text{Price of Y}.$$

Option B meets the definition of equimarginal principle as the key word here is **each** that could be more than one commodity.

Why are other options wrong?

A- the successive units of the same products leads to the law of diminishing MU not the equal amount. This is because an additional unit gives lesser satisfaction than the previous ones

C- the total utility would be changed as more additional units consumed

D- MU keeps changing if additional units consumed not remain constant

2-C:

A natural monopoly produces in an industry where one firm can supply the entire market at a lower long-run average cost than multiple competing firms. This is typically due to vast economies of scale and high fixed costs associated with extensive infrastructure.

If unregulated, a natural monopoly will produce at the quantity where marginal cost equals marginal revenue ($MC=MR$), and then charge the highest price consumers are willing to pay at that output level that is P_1 in the diagram. But this point will not achieve either allocative and productive efficiency.

Allocative efficiency is where $MC=AR$ that is the socially optimal output

Productive efficiency is where ATC is minimised at the given level of output

As we can see in the diagram, the P_3 and Q_3 will meet the condition of the allocative efficiency ($MC=AR$). However, natural monopolies can't be productively efficient as it does not meet its condition firstly. Also, due to lack of competition, which reduces the incentive for efficiency and innovation.

Why are other options wrong?

A- At P_2 and Q_2 , AC equals AR that meets the condition of normal profit in the long run.

B- At P_2 and Q_1 , MC equals MR that meets the condition of the profit maximising level of output.

3-A:

A budget line is a graphical representation of all possible combinations of two goods a consumer can purchase given a fixed income and the prices of those goods.

A change in consumer preferences or a change in the relative prices of both goods that happens at the same time as an equal proportional change in income will not affect the budget line. This is because a budget line represents the combinations of goods a consumer can afford, and these factors would not change the purchasing power or the affordability of the goods. Therefore, option A makes the correct choice.

Why are other options wrong?

B- change in level of income taxes would likely affect the consumers income and purchasing power and therefore this would affect the budget line respectively.

C- the change in prices of one good will cause the budget line to rotate. If prices of one good increases, the budget line pivots inward on the axis of that good. If it decreases, the line pivots outward. Thus affecting the budget line.

D- changes in incomes will cause shifts in the budget line. If income rises, the budget line shifts outward and if income falls, the budget line will shift inward. Thus affecting the budget line.

4-C:

Consumers initially operate where their perceived marginal private costs (MPC) equal their perceived marginal private benefits (MPB), which is the intersection of the MPC1 and MPB1 curves. This occurs at quantity Q1.

The consumer realizes they have underestimated the MPC, meaning the true costs are higher (represented by the MPC2 curve). However, their perceived MPB remains at MPB1.

The new, more accurate equilibrium for the consumer, based on their correct cost perception and current benefit perception, would be at the intersection of the true MPC2 and MPB1 curves. This intersection occurs at quantity Q3.

The consumer was consuming at Q1 but should be consuming at Q3 to be efficient with their new understanding of costs. Since Q1 is greater than Q3, the consumer is consuming more than the optimal amount. Therefore, the consumer is overconsuming.

The extent of the overconsumption is the difference between the actual consumption Q1 and the optimal consumption Q3 which is the quantity Q1Q3.

Why other options are incorrect:

A and B: The consumer is overconsuming, not under-consuming.

D: The quantity Q4 relates to the intersection of MPC1 and MPB2 which is a different scenario not directly relevant to the consumer realizing their MPC underestimation while MPB remains at MPB1.

5-C:

Internal economies of scale are cost advantages that arise from within a single firm as it increases its size and output, leading to a lower average cost per unit. These advantages come from factors like bulk purchasing discounts, the use of more efficient technology, the ability to specialize management roles, and lower average costs of marketing and finance.

Risk-bearing is an internal economy of scale where a larger firm's ability to spread costs and risk across a wider range of products, markets, or operations allows it to absorb losses more easily. By diversifying its activities, a large firm can offset potential losses in one area with profits in another, making it more resilient and reducing its overall average cost per unit.

Therefore the option C is an example of an internal economy of scale.

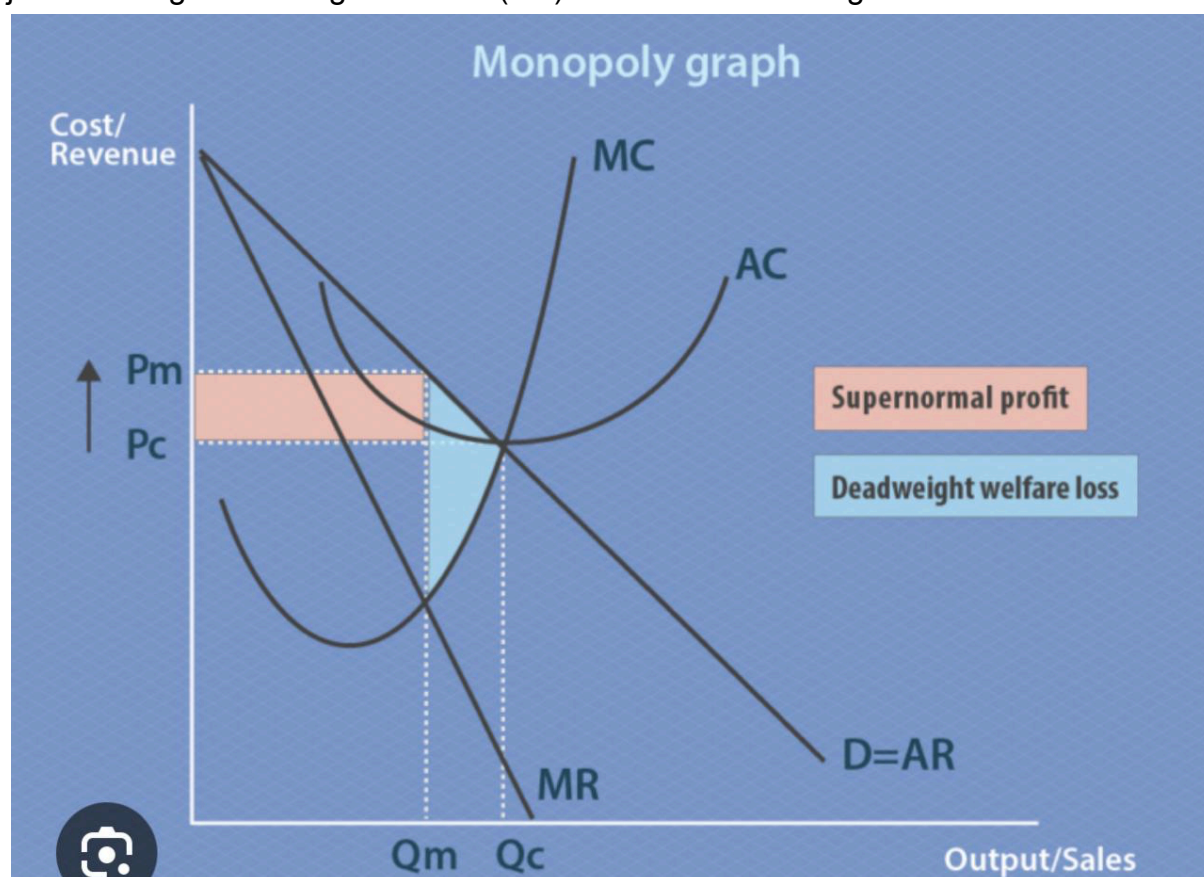
Why are other options wrong?

Other options include the example of an external economies of scale as it includes advantages that benefit all the firms in an industry.

6-D:

Normal profit occurs when price equals average total cost, meaning the firm is breaking even and earning zero economic profit.

A monopoly diagram showing normal profit has the average total cost (ATC) curve just touching the average revenue (AR) as shown in the diagram below.



Why are other options wrong?

A- $MC=AR$, profit maximisation

B- $MR=0$, sales revenue maximisation

C- $MC=AR$, allocative efficiency

7-C:

Small firms are flexible, niche-focused, often owner-operated with less access to capital but strong customer relationships, while large firms leverage economies of scale, advanced tech, broad market reach, and easier financing, leading to more bureaucracy but greater stability and resource access, though they face shareholder pressure. Their economic roles differ: small firms foster innovation and local jobs, while large firms drive industrial output and technological advancement.

8-C:

A negative consumption externality is a cost that a third party bears as a result of someone else's consumption of a good or service. It occurs when the consumption of a product creates a detrimental impact on others, where the private costs of consumption are less than the social costs. Examples include the costs of secondhand smoke from smoking cigarettes or the pollution caused by driving gasoline-powered cars.

To meet the government aim of reducing the negative externalities of consumption, the government should impose **indirect tax** as such imposition would result in an increase in price level that would likely to reduce the quantity demanded by consumers and therefore signals to the producers to reduce the quantity supplied. To reduce the quantity demanded by a greater percentage, the PED should be greater than 1. This is because, in case of elastic demand consumers show greater responsiveness to an increase in price than in inelastic demand.

9-B:

tradable pollution permits, also known as Cap-and-Trade, are market-based tools where a government sets an overall limit (cap) on pollution, issues permits for that amount, and allows companies to buy/sell these permits, creating a market price that incentivizes firms to reduce emissions efficiently by selling surplus permits or buying more when needed. This system internalizes the negative externality of pollution.

A is incorrect because consumers do not directly determine the optimum pollution level; the government sets the overall cap. The market then determines the price of permits.

B is correct because firms that can reduce their pollution below their allocated permit level can sell their unused permits to other firms, thereby receiving a financial reward for their efficiency.

C is incorrect because the government raises revenue from the initial sale of permits, not from the resale between firms in the secondary market.

D is incorrect because the scheme involves significant administrative costs for monitoring emissions, issuing permits, and enforcing compliance.

10-C:

Government failure occurs when a government intervention leads to a net welfare loss or an inefficient allocation of resources compared to the market outcome. In this case, the government built a new highway to reduce traffic congestion, but the result was that car journeys took even longer due to an unintended consequence (more people switching to private cars). The failure is specifically due to the unintended outcome, which is a common form of government failure.

(A) Opportunity costs are always a factor in government decisions, but they are not the direct cause of the failure in this scenario; the failure lies in the outcome itself.

(B) While the wishes of the majority were ignored, this is a political issue, not the economic definition of government failure, which is focused on inefficient outcomes.

(D) The problem describes a government intervention (building a highway), not an over-reliance on the market system.

11-B:

The poverty trap is a situation where an increase in an individual's income results in little or no increase in their net disposable income. This happens because an individual's income results in little or no increase in their net disposable income. This happens because the individual loses a significant amount of means-tested benefits (benefits that are only paid to people with low incomes) or starts paying income tax at a higher rate.

When benefits are means-tested, an individual loses eligibility for these benefits as their income rises. This withdrawal of benefits acts as a very high effective marginal tax rate, discouraging them from seeking higher-paying work or working more hours.

Options A, C, and D would likely help individuals avoid the poverty trap by increasing their net income potential:

(A) national minimum wage is changed: A change in the national minimum wage affects the entry point into employment or the base level of income, but it does not directly create the mechanism of benefits being withdrawn as an individual increases their working hours or earns more.

(C) subsidised housing payments for low-income earners are increased: Increasing these payments provides more support. The trap is caused by the withdrawal of such payments as income rises, not by the increase itself. This change would make the benefits system more generous, but the withdrawal mechanism would still cause a trap if it exists.

(D) the income level at which income tax is first paid is increased: This measure actually helps to reduce the poverty trap. By raising the tax-free allowance, individuals can earn more income before they start paying income tax, thus keeping a larger portion of their increased earnings and providing an incentive to work more.

12-B:

When a minimum wage (MW) is introduced above the equilibrium, the quantity of labour demanded by firms decreases from Y to X. This means some existing jobs are lost. The distance XY represents the number of existing workers made redundant.

At the higher wage, the quantity of labour supplied increases from Y to Z. This means more people are willing to work. The excess supply of labour (unemployment) is the total distance XZ. This total unemployment consists of the existing workers who lost their jobs (XY) and the new people entering the labour market because of the higher wage (YZ). Thus, the distance YZ represents the number of unemployed new entrants.

13-C:

Nudge theory is a concept in behavioural science that suggests indirect suggestions and positive reinforcement can influence the decisions and behaviour of groups or individuals, without forbidding any options.

Options A, B, and D involve significant changes to the system or impose heavy penalties, which are forms of direct enforcement, not "nudges".

Option C, providing information about the undesirable effects, aims to subtly influence taxpayer behaviour by appealing to their sense of social responsibility or fear of negative consequences, thus "nudging" them towards compliance without removing the option to evade tax.

14-B:

Economic rent is the difference between the earnings of a factor of production (labour in this case) and its transfer earnings (the minimum amount required to keep the factor in its current use). It is represented by the area above the supply curve and below the equilibrium wage rate.

In the first diagram, area 1 is above the supply curve (S) and below the equilibrium wage rate.

In the second and third diagrams, the supply curves (S) represent perfectly elastic supply, meaning the entire earnings are transfer earnings, and there is no economic rent (area 3 is the total earnings).

In the fourth diagram, the supply curve (S) represents perfectly inelastic supply, meaning all earnings are economic rent (area 4 is the total earnings).

Therefore, areas 1 and 4.

15-B:

The marginal propensity to save (MPS) is calculated as $1 - MPC$

$$MPS = 1 - 0.6 = 0.4$$

The simple multiplier (k) for an economy with no government or foreign trade is given by $1/MPS$

In 2 sector economy, the multiplier is
 $= 1/MPS = 1/0.4 = 2.5$

The difference between the full employment level of national income and the current equilibrium level is the required increase in income (change in Y)
 $= \$15000 \text{ million} - \$10000 \text{ million} = \5000 million

The change in national income is the multiplier multiplied by the change in investment (change in Investment).

We can rearrange this to find the required change in investment:

$$\text{Multiplier} = \text{change in RNI} / \text{change in Investment}$$

$$\text{Change in Investment} = \text{change in RNI} / \text{Multiplier}$$
$$\$5000 / 2.5 = \$2,000.00$$

The investment would have to increase by \$2000 million, which corresponds to option B.

16-C:

Hysteresis unemployment refers to the idea that a period of unemployment can cause permanent damage to a person's employment prospects, leading to long-term unemployment even after the economy recovers. This happens because the long duration of unemployment leads to a loss of job-specific skills, a decline in general work habits, and a reduction in the motivation to work, as described in option C.

Here is why the other options are incorrect:

(A) people who become temporarily unemployed because it takes a short period of time to find a job after they leave school: This describes frictional unemployment, which is the natural, short-term unemployment that occurs as people move between jobs or enter the workforce for the first time.

(B) people who become unemployed when there is a recession but who will find employment as the economy comes out of recession: This describes cyclical unemployment (or demand-deficient unemployment), which is caused by a general lack of demand for labor during a business cycle recession. This type of unemployment is temporary and resolves with economic recovery.

(D) people who become unemployed when established firms close and new firms are created as technology advances: This describes structural unemployment, which is caused by a mismatch between the skills workers have and the skills needed for the available jobs, often due to technological changes or shifts in industry structure.

17-B:

A positive output gap means actual GDP is greater than potential GDP.

So the economy is operating above its long-run capacity.

That tells us two key things:

1. Economic growth is above trend

- If output is above potential, the economy must be growing faster than its long-run (trend) rate.
- Resources are being used very intensively.

2. The general price level is rising

- High demand pushes up wages and production costs.
- This creates inflationary pressure.
- So prices rise.

Why A is wrong

Above trend growth + falling prices

If growth is above trend, demand is strong — prices would not fall. Falling prices happen in recessions (negative output gap).

Why C is wrong

Below trend growth + falling prices

Below trend growth happens when the economy is weak (negative output gap), not positive.

Why D is wrong

Below trend growth + rising prices

Below trend growth contradicts the idea of a positive output gap.

If output is above potential, growth cannot be below trend.

18-A:

Education and training are factors affecting the occupational mobility of labour because they determine the skills and qualifications a worker possesses. A person with a specific set of skills may find it difficult to move to an occupation that requires different skills without further training.

Immigration controls (B), price of housing (C), and transport infrastructure (D) are generally considered factors affecting geographical mobility of labour (the ability of workers to move to different locations for work), not occupational mobility (the ability of workers to change their occupation).

19-A:

When a government sets interest rates artificially low, it encourages borrowing and spending by both consumers and businesses.

Increased Borrowing and Spending: Lower interest rates make it cheaper to take out loans (for houses, cars, business investments, etc.). This increases the overall demand for goods and services in the economy.

Increased Investment (Option C): While increased investment is a consequence, it is generally considered a positive consequence (leading to growth), not a negative one, unless it causes significant overheating or asset bubbles.

Rapid GDP Growth (Option B): Similar to investment, rapid GDP growth is generally a positive consequence, not a negative one.

Reduced Borrowing (Option D): This is incorrect; low interest rates encourage borrowing.

Inflation (Option A): The negative consequence is that this high demand can outpace the economy's ability to produce goods and services, leading to upward pressure on prices, resulting in a higher rate of consumer price inflation.

20-C:

The quantity theory of money is expressed by the equation of exchange: $MV=PT$, where (M) is the money supply, (V) is the velocity of circulation, (P) is the general price level, and (T) is the total number of transactions or (Y) for real output).

For the general level of prices (P) to remain unchanged, the product of (M) and (V) must change by the same percentage as (T). In option C, the money supply (M) rises by 3% and the total number of transactions (T) rises by 3%, while the velocity of circulation (V) remains unchanged.

The left side of the equation (MV) increases by 3% (since M) increases by 3% and (V) is constant). The right side of the equation (PT) also increases by 3% (since T) increases by 3% and (P) is constant).

Therefore, the equation remains balanced, and the general level of prices (P) remains unchanged.

21-D:

An inflation rate is rising from 3% to 5%, which indicates that AD starts to rise in an economy. A rise in AD would likely bring the multiplier effect and thus reduce the unemployment level. Also inflation and unemployment has an inverse relationship, this means when inflation rate is increasing, the unemployment rate will be falling.

22-A:

Macroeconomic policy objectives are typically broad goals that affect the entire economy, such as economic growth, price stability, and low unemployment.

A) to improve sustainability: Sustainable economic growth is a key macroeconomic objective for developed economies, aiming to meet current needs without compromising future generations' ability to meet theirs.

B) to provide public goods: This is a microeconomic function of the government, addressing market failure for specific goods (like streetlights or national defense).

C) to reduce the power of trade unions: This is a structural or microeconomic reform, not a primary, universal macroeconomic policy objective.

D) to subsidise the electricity supply industry: This is a specific, microeconomic intervention in a single industry, not a broad macroeconomic goal.

23-A:

If inflation expectations are unchanged, an increase in government spending would stimulate a multiplier effect, thus affecting the price, output and employment level in the economy. This suggests that unemployment would fall but at the expense of higher inflation rate. Therefore, it conflicts with the government objective of low inflation.

B- if inflation expectations fall, then higher government spending wouldn't bring a bigger multiplier effect in an economy. Therefore low and stable inflation rates can be maintained.

C- labor productivity would likely reduce unemployment and control inflation at the same time. This is because it increases the AS, reducing price level and increasing real output.

D- if labour supply increases, this also suggests that a larger number of workers can be hired at a lower cost by firms. This reduces the level of unemployment and also cost-push inflation is also controlled.

24-D:

External value of money: The external value of a currency is its exchange rate against other currencies. When an exchange rate depreciates, it means the currency is worth less in terms of foreign currency. Therefore, the external value of money falls.

Internal value of money: The internal value of money refers to its purchasing power within the economy. Since the economy imports a large proportion of its raw materials, a depreciation in the exchange rate makes these imported raw materials more expensive in local currency. This increases the cost of production for businesses, leading to an increase in the general price level (inflation). As prices rise, the purchasing power of the money decreases. Therefore, the internal value of money falls.

25-B:

When a government imposes a tariff, it protects domestic producers from foreign competition by making imported goods more expensive. This allows less efficient domestic firms to survive and operate at a higher cost than international competitors, leading to productive inefficiency in the country's steel industry.

A is incorrect because the tariff raises the domestic price of steel, making the country's steel exports less price competitive in the global market.

C is incorrect; while a large country might improve its terms of trade, generally, tariffs can invite retaliation and disrupt trade, which doesn't guarantee a permanent improvement.

D is incorrect as the tariff increases the price and quantity supplied by domestic producers, thus increasing their producer surplus.

26-C:

The Human Development Index (HDI) is a composite statistic of three key dimensions of human development: health, education, and standard of living. Each dimension is measured by specific indicators:

- Health is measured by life expectancy at birth.

- Education is assessed through average years of schooling (for adults) and expected years of schooling (for children).

-Standard of living is evaluated using Gross National Income (GNI) per capita.

Infant mortality rate is an important health indicator but is not directly included in the calculation of the HDI.

27-B:

A revaluation (currency appreciation) and its effect on the current account over time. The graph shows:

- * Initially: current account improves slightly
- * Then: it worsens significantly over time
- * This is the reverse J-curve effect.

Key concept: Marshall–Lerner condition

For a revaluation (appreciation):

- Exports become more expensive.
- Imports become cheaper.

In the short run , the demand and supply of exports and imports is inelastic. Therefore in Option B the sum of $PED_x + PED_m$ is less than 1 that is $(0.4+0.4 = 0.8)$

If demand is inelastic in the short run, quantities don't change much at first. Therefore the current account improves and continues to rise.

In the long run, the demand and supply of exports and imports is relatively elastic. Hence in option B the sum of $PED_x + PED_m$ is greater than 1 that is $(0.8+0.8 = 1.6)$ Therefore the current account worsens and continues to fall.

28-C:

The International Monetary Fund (IMF) primarily focuses on macroeconomic stability, exchange rates, and balance of payments issues. It provides financial assistance to countries experiencing economic crises or balance of payments difficulties, often with conditions for policy adjustments.

(A) Encouraging exchange rate stability is a key function of the IMF.

(B) Providing financial assistance for a country with failed economic policies (leading to a crisis) is a primary role.

(D) Providing loans for countries following a natural disaster can be part of the IMF's function if it impacts the country's balance of payments and overall economic stability.

(C) Providing funds for specific infrastructure projects like a water purifying plant is typically the role of development organizations like the World Bank, not the IMF. The IMF deals with broader economic issues, not specific project financing

29-D:

The total net gain to the domestic economy when a tariff is removed (moving from the world price with tariff, P_1 to the world price without tariff, P_2) is the reduction in deadweight loss.

Before the tariff is removed, the deadweight losses are the areas labeled w and y . Area w represents the deadweight loss from inefficient domestic production (production distortion loss). Area y represents the deadweight loss from reduced consumption due to the higher price (consumption distortion loss). When the tariff is removed, the price falls to P_2 , the domestic production decreases, and domestic consumption increases. The removal of the tariff eliminates these two deadweight losses, which become the net gain to the economy

30-B:

Economic development requires flexibility in resources and labor to move towards more productive industries. Increased government legislation to protect existing employment (option B) often makes it difficult for firms to fire workers or restructure, leading to labor market rigidity. This prevents resources from moving to more efficient sectors, thus hindering economic growth and development.

Options A, C, and D generally promote economic growth by increasing resource efficiency, encouraging innovation, or expanding export markets.

